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| **Fiscal Year:** | FY 2008 | **Task Last Updated:** | FY 06/05/2009 |
| **PI Name:** | Orasanu, Judith M. Ph.D. | | |
| **Project Title:** | Enhancing Team Performance for Exploration Missions | | |
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| **Program/Discipline--Element/Subdiscipline:** | NSBRI--Neurobehavioral and Psychosocial Factors Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
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| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
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| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Kraft, Norbert ( NASA ARC ) | | |
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| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | AIMS: Since the beginning of human space flight, psychological and interpersonal aspects of crew behavior have jeopardized crew health, safety and mission success. The goal of this project is to optimize team psychosocial adaptation and performance in exploration missions (Risk #24, NASA Critical Path Roadmap, and BHP Team Risk). We aim to enhance the ability of crews to respond adaptively to unanticipated problems, to maintain positive social and task cohesion, and to collaborate effectively under stress. The research will (a) develop and validate tools for monitoring team cohesion; (b) determine the effects of team composition and task and interpersonal stressors on team performance and cohesion; (c) determine individual differences in response to stressors; (d) identify effective team interaction strategies for managing stress; and (e) develop and validate a training protocol to assure effective team performance and high cohesion.<br>APPROACH: A teamwork simulation laboratory has been created to address the above goals. Teams consisting of 4 or 5 | | |

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| **Task Description:** | members (one of whom is assigned the role of Base Coordinator, BC) engage in Antarctic or Lunar search missions over 3 days. Social and task stressors, crew composition, and team training protocols have been manipulated to determine their effects on team processes, cohesion and performance. Time-synchronized performance, video, and physiological data, along with questionnaires tapping workload, team dynamics, personality, and other individual characteristics, support analysis of team processes and outcomes.<br><br>The Group Diagramming Method (GDM) and Linguistic Analysis of team functioning (LA). Using data from three experiments we established the construct validity of these measures and determined their concurrent and predictive validities in relation to team performance.<br><br>GDM. Team cohesion is typically assessed using self-report measures such as the Group Environment Scale (GES) in which team members rate their team environment as a whole. In contrast, GDM characterizes group interactions along three dimensions (positive/negative; task-oriented/expressive; dominant/submissive) and provides a graphical representation of team climate. GDM was validated by comparison with the GES and ratings of social and task cohesion.<br><br>Team interaction behaviors were stable across GDM administrations at the end of mission Days 1 (T1) and 3 (T2), indicating good test-retest reliability and stability of group measures. Cohesion in the GDM is represented by clustering or low dispersion of team member scores on the group diagram plane (positive/negative and expressive/task-oriented dimensions) and a team Center of Gravity (COG) on the positive side of the group diagram. High ratings on the GES cohesion subscale were associated with the following GDM measures: an expressive team COG, equal team member participation, and a positive and expressive BC. In contrast, high levels of GES anger and aggression were associated with high dispersion of team scores and a negative and task-oriented COG. Similarly, high ratings on the social and task cohesion scales were associated with strong clustering of team scores and a positive team COG. The social cohesion scale was also associated with an expressive COG. Dispersion scores on both the task and social cohesion rating scales were highly correlated with dispersion scores on all three GDM dimensions.<br><br>LA: Analysis of team communication involved three measures of cohesion: the inclusiveness and symmetry of team members' interactions, and expressed affect. Interactions in cooperative teams involved more team members, indicating that these teams were more inclusive than competitive teams. Cooperation was also associated with balanced interactions and equal participation. Cooperative teams showed significantly more positive affect than competitive teams on Day 3, but no difference between groups was observed on Day 1. The extent to which team interactions were positive, symmetric and inclusive was also related to members' perceptions of their team's communication efficacy and GES cohesion: the more symmetric and balanced their interactions, the higher team members rated their communication efficacy and cohesion. Fewer positive interactions led to lower efficacy and cohesion ratings.<br><br>The high correlations between alternative methods of assessing cohesion support the validity of both the GDM and LA approaches.<br><br>Cohesion Measures and Performance: Both the LA and GDM cohesion measures were related to team performance. Using the GDM, an expressive team COG and more equal participation correlated with higher performance as measured by missions completed, average mission score, and higher levels of cooperation. LA communication analyses of high and low performing teams were consistent with GDM findings. Successful teams had more symmetric, inclusive, and collaborative interactions, and team members were more positive and supportive. No team cohesion measures at T1 predicted team performance at T2 except for BC behaviors: lower BC dominance and higher expressiveness predicted more mission tasks attempted. However, team performance at T1 predicted cohesion at T2: number of missions completed and emergency task points predicted team positive COG and more equal participation rates, respectively.<br><br>IMPLICATIONS: Tools for monitoring crew cohesion will become more essential during long-duration space missions, when the risk of psychosocial problems increases. Communication delays and disruptions mean the crew will function more autonomously than at present. Providing them with valid and trusted diagnostic feedback concerning their interactions will be important for self-management of interpersonal problems. The monitoring tools validated in this study are one step toward developing automated crew self-monitoring systems. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | A substantial portion of work in industry, academe, and government is conducted in teams. Better understanding of factors that contribute to effective and cohesive team performance will have benefits in many domains here on earth, in addition to payoffs for space crews. These benefits will be especially important in high-risk industries such as medicine, nuclear power, chemical processing, aviation, and the military. Potential benefits include greater efficiency, fewer errors, greater productivity, and less time lost due to interpersonal friction or stress in the workplace.<br>The instruments that we are developing for assessing team cohesion and communication can help design work teams that are highly cohesive and productive. These instruments will also be useful for monitoring changes in team dynamics over time and for providing theoretically-grounded recommendations to improve team cohesion and performance. Ultimately, these tools will be automated and will give feedback in real time. Finally, this research will produce validated team training protocols that could be built into education programs in colleges, professional schools, medical team training, and preparation for crisis-management teams (e.g., negotiators, firefighters, police, or disaster relief teams). These protocols include team self-evaluation and debriefing skills that are helpful in all teams, but essential for maintaining adaptive behaviors in teams that operate in high-stress, high-risk dynamic environments. |
| | Major tasks accomplished during Year 4 addressed BHP Team Gap 2: What are optimal ways to monitor and measure crew cohesion, crew performance, and crew-ground interaction for exploration missions? Activities included analysis of data from studies 2 (S2) and 3 (S3) to establish construct validity of two new cohesion monitoring tools and to determine the concurrent and predictive validities of the cohesion measures with respect to team performance.<br>S2 examined the effects of two between-subject variables on team performance: team stress (competitive vs. cooperative team goals) and team gender composition (all male, all female or 2 males/2 females). S3 examined the effects of two team training protocols on team performance and cohesion: Interpersonal Relations Training (IRT) designed to build trust, self-understanding, and team cohesion; and Team Adaptation and Coordination Training (TACT) focused on information sharing, planning, team coordination/backup, and self-evaluation. Two control conditions were used: Team |

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| **Task Progress:** | Familiarity Control (FC) involving live action role play but no training; and No-training/No-familiarity (Control).<br><br>In S2, four-member crews were randomly assigned to competitive or cooperative goal conditions (24 crews, N = 96). Three crew members conducted simulated geological searches in Antarctica, supported by a base coordinator (BC) who evaluated and shared task-relevant satellite information and helped crews refuel and resupply. In S3, five-member gender-mixed crews (2 F/3 M) were randomly assigned to one of the 4 experimental conditions (24 crews, N = 120). Four crew members searched for water on the lunar surface, supported by a BC. All teams performed six 75-minute simulated missions over three days (1 difficult/1 moderate per day).<br><br>The two new approaches to monitoring team cohesion are the Group Diagramming Method (GDM) and Linguistic Analysis (LA). GDM involves team members rating each other on 26 behavioral adjectives that map onto three dimensions: Positive vs. Negative, Task-oriented vs. Expressive, and Dominant vs. Submissive. Statistical analysis of the ratings yields a graphic representation of the team's interaction patterns. Ratings occurred at the end of Days 1 and 3 to permit assessment of changes over time.<br><br>LA involves evaluating social aspects of crews' communication in relation to goal conditions. Based on our previous research, we developed a new procedure to code relational aspects of communication directly from videotapes without prior transcription. This procedure was applied to team interactions during missions on Days 1 and 3 for 12 of the 24 crews in S2. Analyses focused on the inclusiveness of team members' interactions, conversational control, and positive affect.<br><br>An observer rated S2 crew interactions using the GDM behavioral checklist for the same 12 crews whose communication we coded with LA. When analyses are completed, this will permit comparison of cohesion assessed by the two methods. |
| **Bibliography Type:** | Description: (Last Updated: 05/21/2010) |
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